

Passive Risk Detection Report

devdaytest1 - Harness CD - Cluster: validation-cluster - Namespace: hce

Scan Identity: test-scan-1778747446 - Scanned 2026-05-14

HARNESS RESILIENCE TESTING - EXPERIMENTAL | Confidential

58/1000

PIPELINE RISK SCORE

5

HIGH RISKS

2

MEDIUM RISKS

0

LOW RISKS

7

TOTAL RISKS

1 Executive Summary

This report presents the results of a passive resilience risk scan performed by the Harness Resilience Testing engine against the devdaytest1 project. The scan targeted the Kubernetes cluster 'validation-cluster' in namespace 'hce'. A total of 1 service(s) were identified and evaluated for resilience risks without requiring any code changes, instrumentation, or manual scan initiation.

The scan detected 7 risk(s) across 1 service(s): 5 High, 2 Medium, and 0 Low severity. All 7 identified risks are currently Open. The pipeline risk score is 58/1000. High-severity risks include missing resource limits, single-replica deployments, missing health probes, missing network policies, and untagged container images - all posing direct outage or security exposure.

Coverage gaps are critical: no chaos experiments and no load tests exist for any scanned service. All passive risks remain unconfirmed and no fault-tolerance baseline has been established. Immediate action is recommended to remediate High-severity findings and establish a resilience testing baseline before the next production deployment.

Scan Metrics

Metric	Detail
Service Count	1
Total Risks	7
Services Affected	1
Chaos Coverage	0% (no chaos tests)
Open Risks	7
Scan Type	kubernetes-security

2 Services Found in Scan

All 1 service(s) detected in the scan were passively evaluated - no code changes, instrumentation, or manual configuration required.

Service	Total Risks	High	Medium	Low	RRS	Categories
nginx	7	5	2	0	58	Availability, Reliability, S

* RRS = Resilience Risk Score: (High x 10) + (Medium x 4) + (Low x 1). Sorted by RRS descending.

3 Risks Found

7 risks detected across 1 service(s) - 5 High, 2 Medium, 0 Low.

HIGH

K8S Missing Resource Limits

k8s-missing-resource-limits::deployment/nginx::hce

The nginx container has no CPU or memory limits. It can consume all available node resources, starving sibling pods.

Affects: Deployment/nginx in namespace hce

Unbounded resource consumption causes noisy-neighbour failures on the node. Memory hog can trigger kernel OOM kills on sibling pods.

First Detected: 2026-05-14 10:21 UTC Last Detected: 2026-05-14 10:21 UTC

KUBERNETES RELIABILITY NGINX

HIGH

K8S Single Replica

k8s-single-replica::deployment/nginx::hce

The nginx Deployment runs with replicas: 1, making it a single point of failure. A pod-delete fault causes 100% service unavailability until the pod restarts.

Affects: Deployment/nginx in namespace hce

Complete service downtime during any pod disruption, node maintenance, or OOM kill. No redundancy to absorb pod-level failures.

First Detected: 2026-05-14 10:18 UTC Last Detected: 2026-05-14 10:18 UTC

KUBERNETES AVAILABILITY NGINX

MEDIUM

K8S Missing Resource Requests

k8s-missing-resource-requests::deployment/nginx::hce

The nginx container has no CPU or memory requests. It is assigned BestEffort QoS and will be the first pod evicted under any node pressure.

Affects: Deployment/nginx in namespace hce

BestEffort QoS means nginx is evicted before any other pod on the node when memory or CPU pressure occurs - even if nginx itself is not causing the...

First Detected: 2026-05-14 10:22 UTC Last Detected: 2026-05-14 10:22 UTC

KUBERNETES RELIABILITY NGINX

HIGH

K8S Missing Probes

k8s-missing-probes::deployment/nginx::hce

The nginx container has no livenessProbe or readinessProbe. Kubernetes routes traffic to unhealthy pods and never restarts stuck ones.

Affects: Deployment/nginx in namespace hce

Traffic is sent to pods that are restarting or deadlocked. Without livenessProbe, a hung nginx process is never restarted automatically.

First Detected: 2026-05-14 10:22 UTC Last Detected: 2026-05-14 10:22 UTC

KUBERNETES AVAILABILITY NGINX

HIGH

K8S Missing Network Policy

k8s-missing-network-policy::deployment/nginx::hce

The nginx Deployment has no NetworkPolicy. All ingress and egress traffic is unrestricted, enabling lateral movement from any compromised pod in the cluster ...

Affects: Deployment/nginx in namespace hce

Any pod in any namespace can send traffic to nginx without restriction. A compromised workload can exfiltrate data, probe nginx endpoints, or use i...

First Detected: 2026-05-14 15:14 UTC Last Detected: 2026-05-14 15:14 UTC

KUBERNETES SECURITY NETWORK NGINX

MEDIUM

K8S Missing Pod Anti Affinity

k8s-missing-pod-anti-affinity::deployment/nginx::hce

The nginx Deployment has no podAntiAffinity or topologySpreadConstraints. All replicas may be scheduled on the same node, turning the multi-replica deploymen...

Affects: Deployment/nginx in namespace hce

Despite having replicas configured, all pods could land on the same node. A single node-drain or node-restart takes down every replica simultaneous...

First Detected: 2026-05-14 10:33 UTC Last Detected: 2026-05-14 10:33 UTC

KUBERNETES AVAILABILITY NGINX

HIGH

K8S Latest Image Tag

k8s-latest-image-tag::deployment/nginx::hce

The nginx container uses image 'nginx' with no tag. A pod restart may pull a different version from the registry, causing silent version drift between replicas.

Affects: Deployment/nginx in namespace hce

Non-deterministic deployments - two pods in the same Deployment can silently run different image versions after independent restarts.

First Detected: 2026-05-14 10:19 UTC Last Detected: 2026-05-14 10:19 UTC

KUBERNETES RELIABILITY NGINX

4 Severity Distribution and Category Breakdown

Severity Level	Count	% of Total	Definition
HIGH	5	71%	High-confidence failure mode with direct outage potential
MEDIUM	2	29%	Elevated risk requiring near-term attention
LOW	0	0%	Low-severity observation or best-practice deviation
TOTAL	7	100%	Across all 1 services

Risk Count by Category

Category	Count	Risks Included
Reliability	3	K8S Missing Resource Requests, K8S Latest Image Tag, K8S Missing Resource Limits
Availability	3	K8S Missing Pod Anti Affinity, K8S Missing Probes, K8S Single Replica
Security	1	K8S Missing Network Policy

5 Recommendations and Next Steps

Priority	Service	Action	Experiment Type	Severity
P1	nginx	Define CPU/memory limits in container spec	Memory Stress	HIGH
P1	nginx	Scale to >= 2 replicas, add PDB, run pod-delete chaos	Pod Delete	HIGH
P1	nginx	Add livenessProbe and readinessProbe to container	Pod Delete	HIGH
P1	nginx	Apply restrictive NetworkPolicy (ingress + egress)	Network Chaos	HIGH
P1	nginx	Pin image tag to specific version (e.g. nginx:1.27.0)	Pod Restart	HIGH
P2	nginx	Set CPU/memory requests for accurate scheduling	CPU Stress	MEDIUM
P2	nginx	Add podAntiAffinity or topologySpreadConstraints	Node Drain	MEDIUM

Recommended Immediate Action:

Run pod-delete chaos experiments against all single-replica deployments, starting with nginx which has no PDB and zero test coverage. This will immediately confirm or disprove complete service outage on pod eviction and establish a resilience baseline.

Auto-parameterised Experiment Config:

namespace: hce
blast_radius: 100%
hypothesis: HTTP 200 on service endpoint at p99 < 500ms
duration: 5 min
rollback_trigger: error rate > 2x baseline

6 Test Coverage Gap Analysis

Zero coverage means a service's resilience posture is entirely unverified - all passive risks remain unconfirmed and no steady-state baseline has been established.

Service	Chaos Tests	Load Tests	Coverage %	Notes
nginx	None	None	0%	Critical
SUMMARY	0/1 covered	0/1 covered	0%	Immediate action required on all services

Services with no tests of any kind: nginx - 1 of 1 services have zero test coverage.

7 Appendix and Glossary

Report Generation Methodology

This report was generated by the Harness Resilience Testing engine using passive risk detection (PRD). The engine scans Harness CD metadata, Kubernetes manifests, and deployment configuration without requiring any code changes, agent installation, or manual scan initiation. All findings are derived from static metadata analysis only - no live traffic was affected and no workloads were modified during the scan. The report is produced automatically at the conclusion of the scan run and reflects the state of the target namespace at the time of scan execution.

Risk taxonomy follows the classification defined in the Harness Resilience Testing specification. Severity levels (High, Medium, Low) are assigned based on the confidence of failure mode and the potential blast radius. The Resilience Risk Score (RRS) is a composite metric: $(\text{High} \times 10) + (\text{Medium} \times 4) + (\text{Low} \times 1)$, used to rank services and provide a pipeline-level gate score. Coverage metrics reflect the presence or absence of chaos experiments and load tests anchored to the scanned services. Monetary estimates, where present, are illustrative and not contractual.

Glossary

Term	Definition
RRS	Resilience Risk Score - composite score: $(\text{High} \times 10) + (\text{Medium} \times 4) + (\text{Low} \times 1)$. Used to rank services and gate pipelines.
PDB	PodDisruptionBudget - Kubernetes policy that limits the number of pods that can be simultaneously evicted during maintenance.
HPA	HorizontalPodAutoscaler - Kubernetes controller that automatically scales pod count based on observed CPU/memory or custom metrics.
MTTR	Mean Time To Recovery - average time to restore a service after an incident. Used in exposure calculations.
PRD-Report	Passive Risk Detection Report - an API-driven resilience risk scan anchored to a CD pipeline execution, requiring no instrumentation.
HIGH	High severity - high-confidence failure mode with direct outage potential. Maps to P1 priority remediation.
MEDIUM	Medium severity - elevated risk requiring near-term attention. Maps to P2 priority remediation.
LOW	Low severity - low-severity observation or best-practice deviation. Maps to P3 priority remediation.